

Purpose

This packet documents the Terraform-based AWS environment I created, including the infrastructure components, validation evidence, assumptions, and AI assistance used during the assignment.

The purpose of this project is to create a complete operational environment that can be deployed by running a defined set of Terraform code. This approach makes the environment repeatable and flexible if a second or third environment is needed, because each deployment is built the same way.

Goal

Using Terraform, I built separate .tf files to describe each infrastructure component, including the VPC, EC2 instances, subnets, internet gateway, route tables, and security groups. Keeping each component in its own file makes the configuration easier to update because I can change one part without searching through unrelated code.

Assignment Summary

The objective of this assignment is to become familiar with Infrastructure as Code (IaC). Terraform supports this goal by allowing the environment to be defined in code rather than configured manually in the AWS console. This makes changes easier to manage and helps the configuration be adapted for other cloud environments.

Claim

I created three Amazon EC2 instances in AWS:

- Microsoft Windows Server 2022 application server
- Microsoft SQL Server 2022
- Microsoft SMTP email server

I can use Terraform to build and destroy the complete environment.

Assumptions

1. I chose RabbitMQ as the message broker because it appeared to be the easiest option to use and had straightforward setup instructions.
2. I chose SQL Server Express because this environment will use only a small amount of data and does not require the full capabilities of SQL Server.

Evidence

I included a layout generated by terraform plan to show the complete environment design. The supporting file is named MyPlan.pdf.

Validation

I built and destroyed the environment multiple times to confirm that it deployed correctly and removed all resources when destroyed.

AI Assistance

I used Microsoft Copilot to provide step-by-step guidance for provisioning the instances and configuring IP addressing, security settings, and network resources. Copilot also helped me understand how to create the Terraform files for the environment.

I also used Copilot to help reformat this document.

Ownership

I am responsible for reviewing the AI-generated guidance, implementing and validating the configuration, documenting results, and escalating unresolved technical or security risks.

Documentation Included

- All Terraform .tf files
- Environment Plan